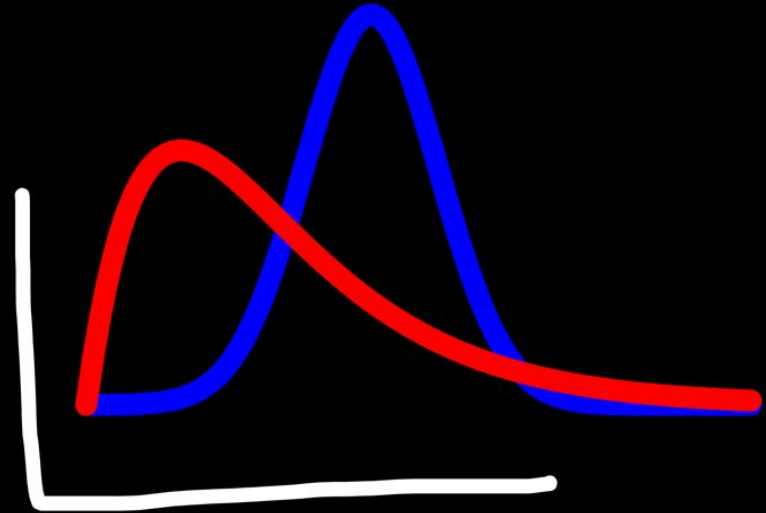


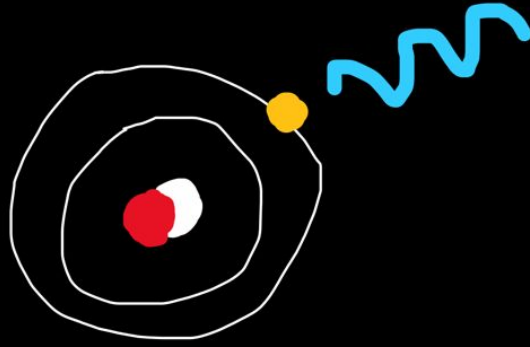
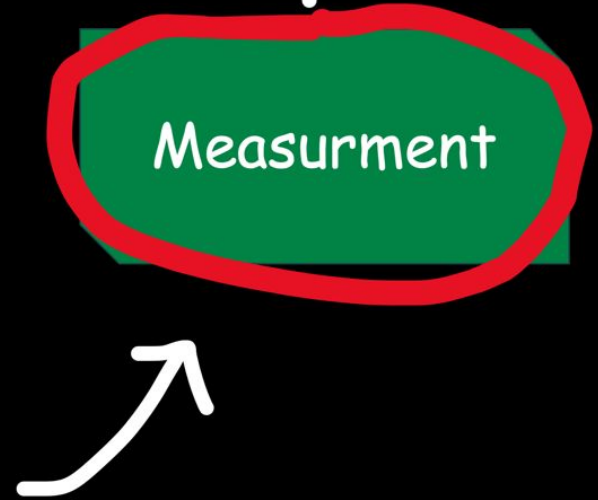
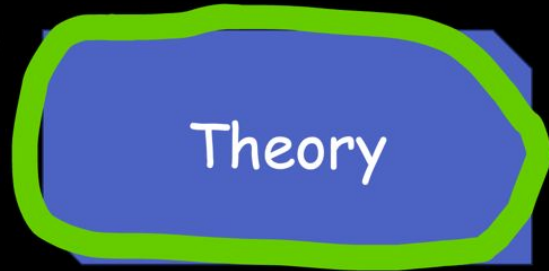
Automated Measurements and Circuit Transfer Functions

Hien D. Luu
Bao N. Gia

Instructor: Sam Condon

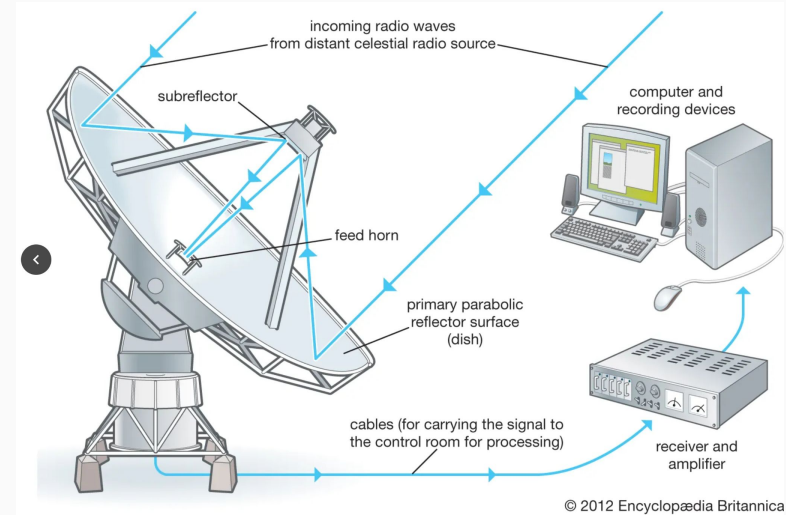


Outline



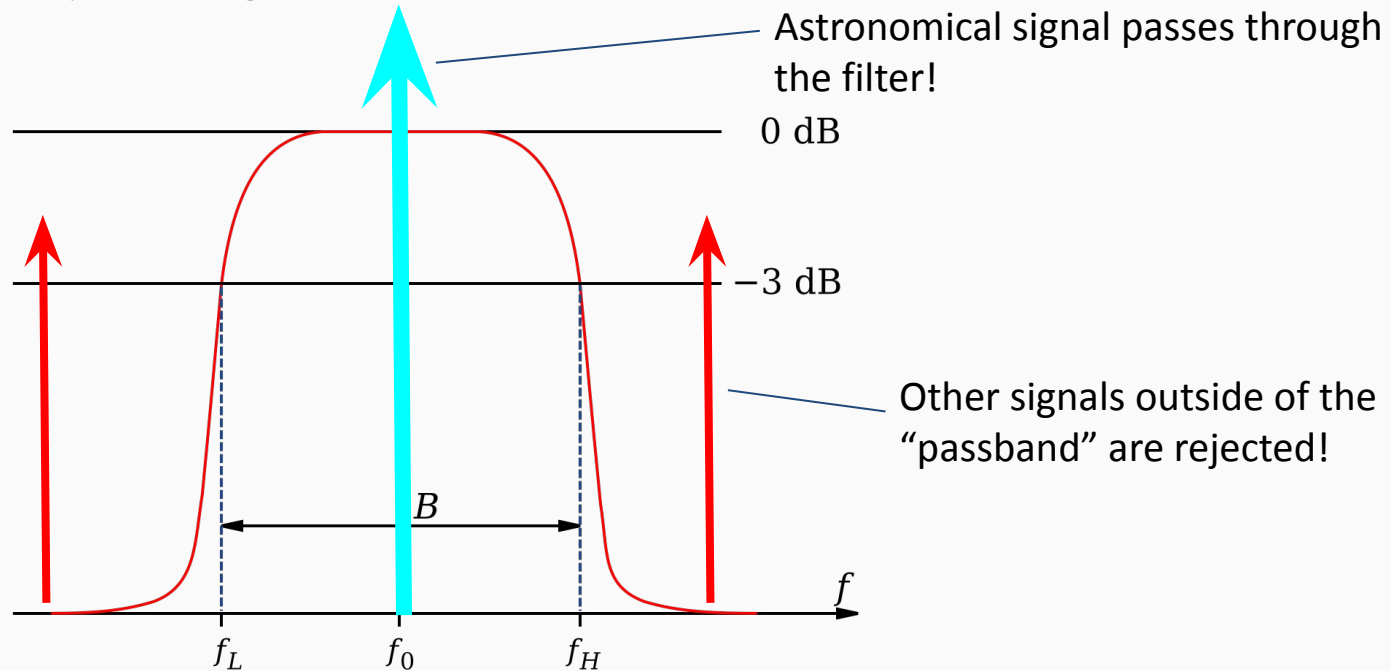
Signal Processing in Astronomy

- Signal processing is an essential task in astronomy
- We need to take light from an astronomical source and convert it into a form that can be stored on our computer



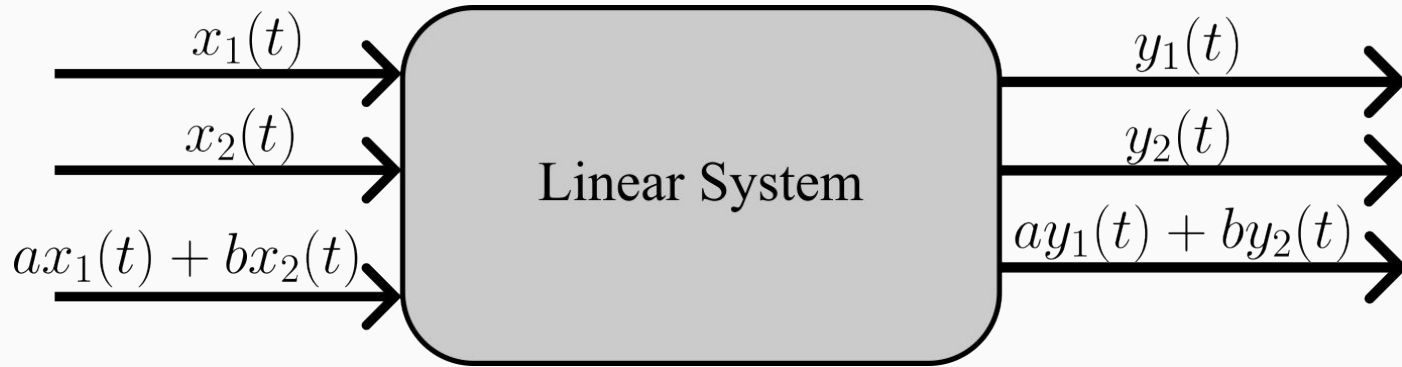
Signal Processing in Astronomy

- One basic example of signal processing in astronomy is the band-pass filter. This filter allows only signals with frequencies within a certain range pass through the filter.
- We use band-pass filters in astronomy to isolate the astronomical signal of interest, before further processing!



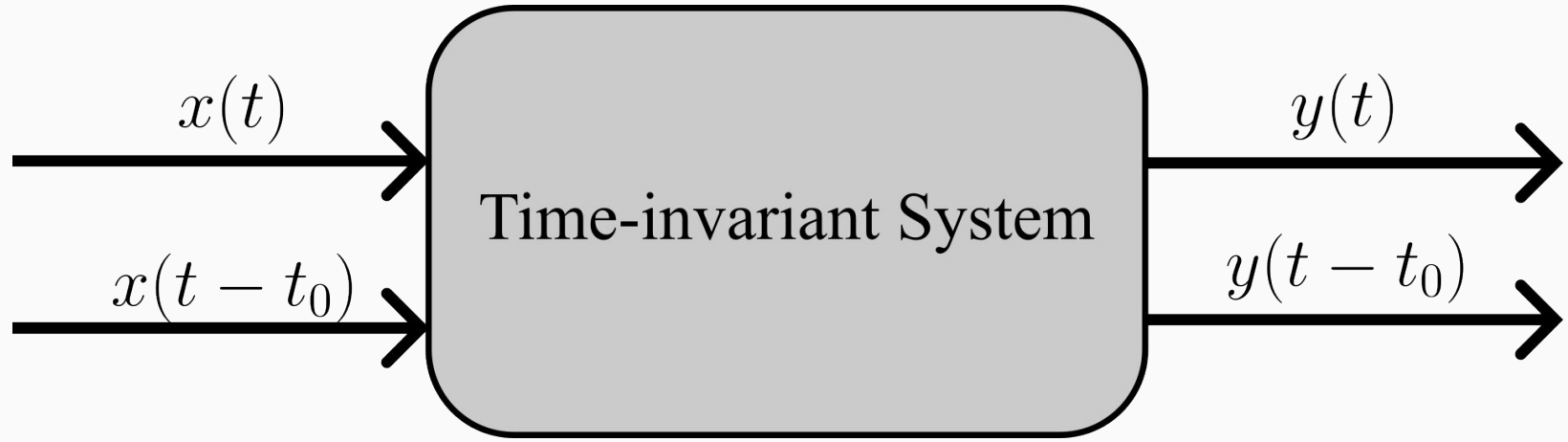
Signal Processing in Astronomy

- Many signal processing systems in astronomy are **linear**:



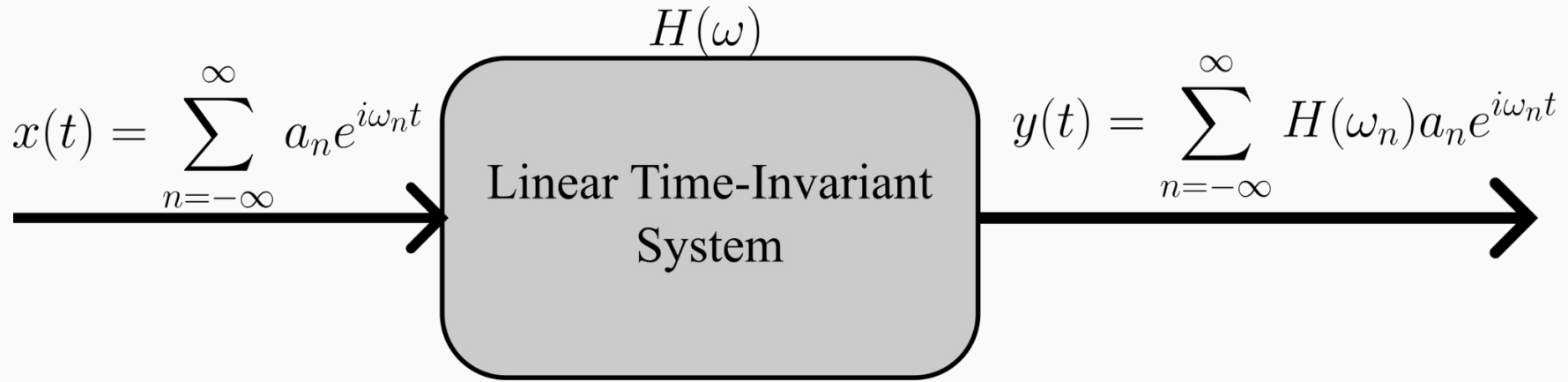
Signal Processing in Astronomy

- Signal processing systems in astronomy are **time-invariant**:



Signal Processing in Astronomy

- Signal processing systems which are **linear** and **time-invariant** can be described entirely by their **Transfer Function**

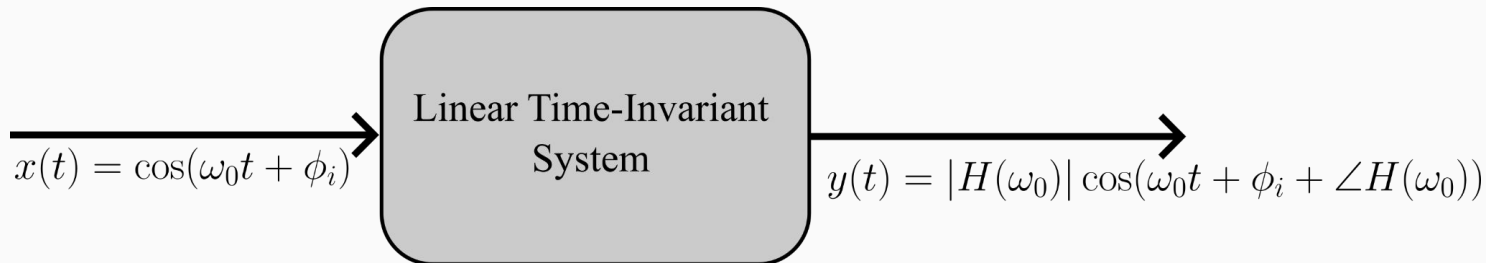


Signal Processing in Astronomy

- The **transfer function** is a complex number with a magnitude and a phase:

$$|H(\omega)| = \sqrt{\mathcal{R}\{H(\omega)\}^2 + \mathcal{I}\{H(\omega)\}^2}$$
$$\angle H(\omega) = \tan^{-1} \left(\frac{\mathcal{I}\{H(\omega)\}}{\mathcal{R}\{H(\omega)\}} \right)$$

- If the input to an LTI system is a cosine, it is scaled by the magnitude and shifted by the phase of the **transfer function**



Signal Processing in Astronomy

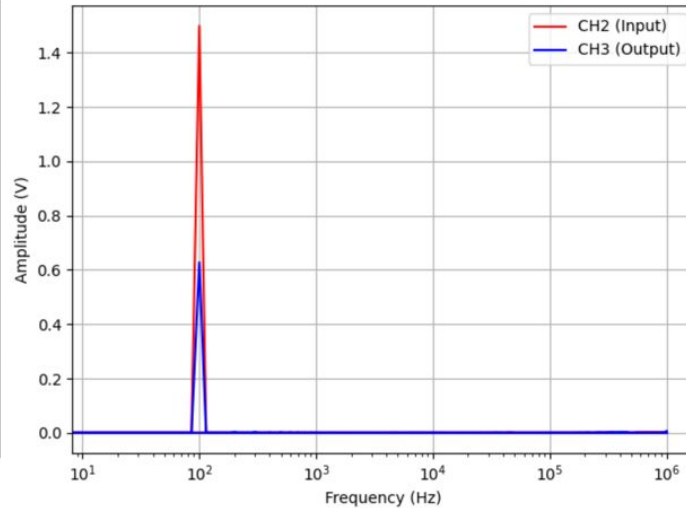
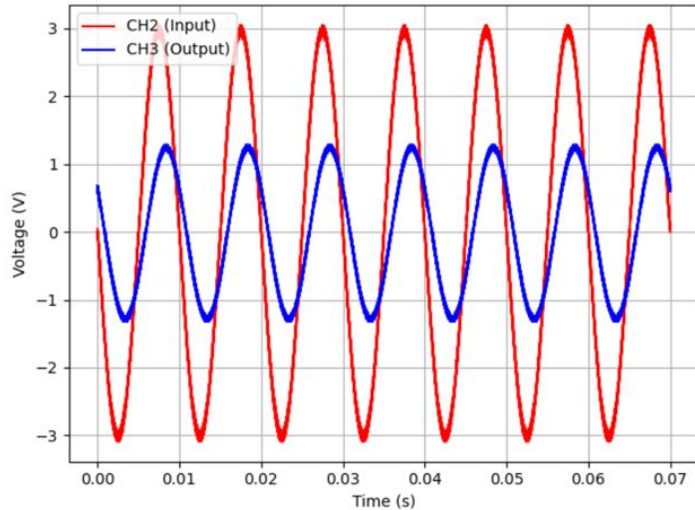
Fourier Transform (FT)

- An integral transform in mathematics
- Convert an input signal from the **time domain** to the **frequency domain** and vice versa
- The result of the Fourier Transform is a complex-valued function of frequency
- **Uncertainty Principle:** A signal localized (concentrated) in the time domain will have a Fourier Transform spread out (dispersed) in the frequency domain, and vice versa

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega$$
$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt$$



1) Theory - Fourier Transform



- + Low pass filter
- + $R = 1 \text{ k}\Omega$
- + $C = 1 \text{ }\mu\text{F}$
- + $f = 100 \text{ Hz}$
- + $V = 3 \text{ V}$

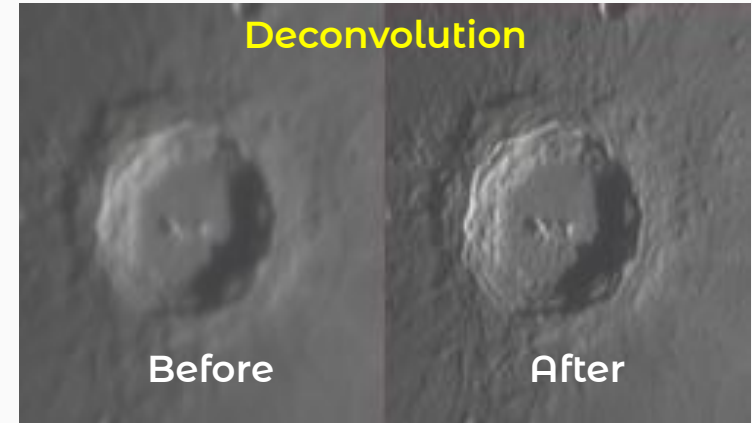
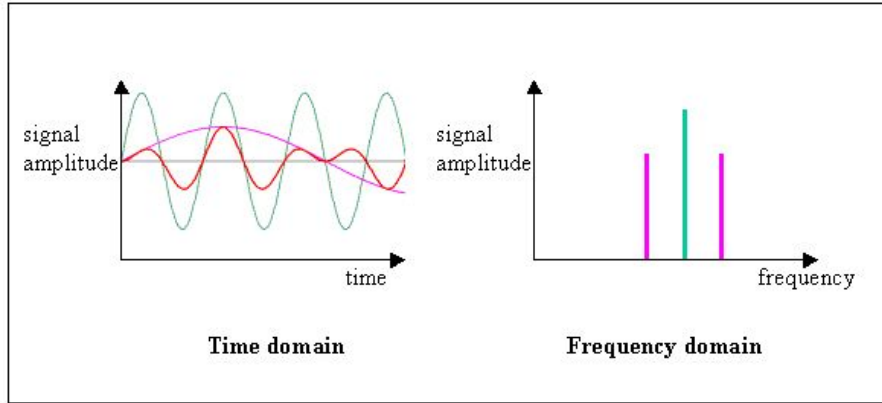
Signal Processing in

Astronomy

The Fourier transform lets us move between the time domain and frequency domain

❖ In the frequency domain we can:

- Easy to isolate and remove noise and unwanted signals → filtering, deconvolution
- Contains important astronomical information → spectroscopy/power spectrums, interferometry



Introduction

❖ Transfer Function (TF)

- A mathematical model of a system's output for any possible input
- Typically reserved for **Linear Time-Invariant (LTI)** systems. The result of the Fourier Transform is a complex-valued function of frequency
- For LTI systems, the TF describes how the system scales and phase-shifts each frequency component of the input signal

Signal Processing in Astronomy

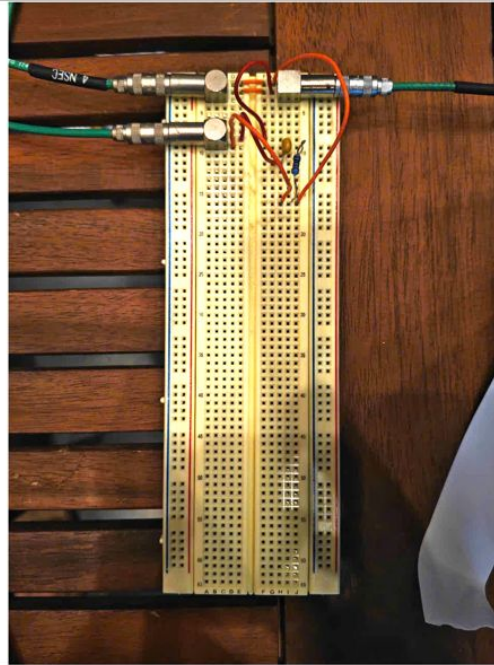
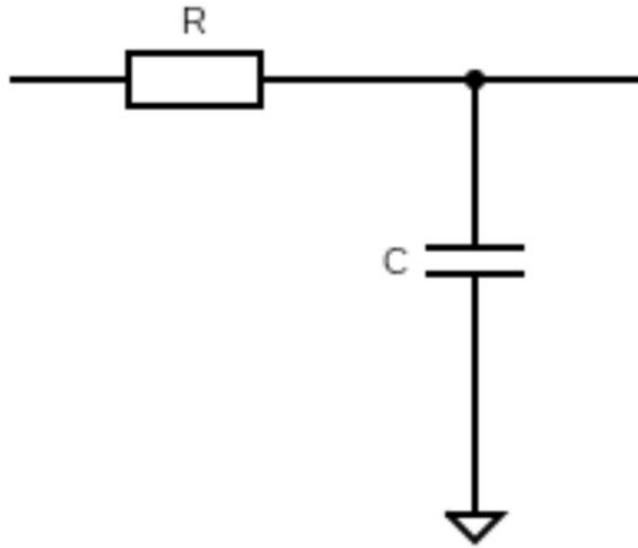
Relationship Between Fourier Transform and Transfer Function

- The Fourier Transform and Transfer Function are foundational tools for frequency-domain system analysis

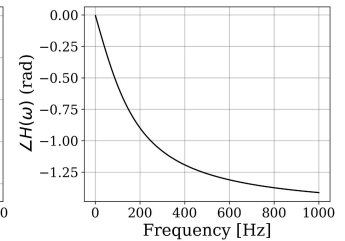
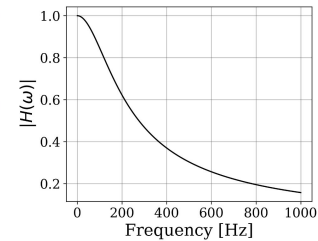
Convolution Theorem:

- In the time domain: Output $y(t)$ is the convolution of input $x(t)$ and impulse response $h(t)$
- In the frequency domain: $Y(\omega) = H(\omega) \cdot X(\omega)$

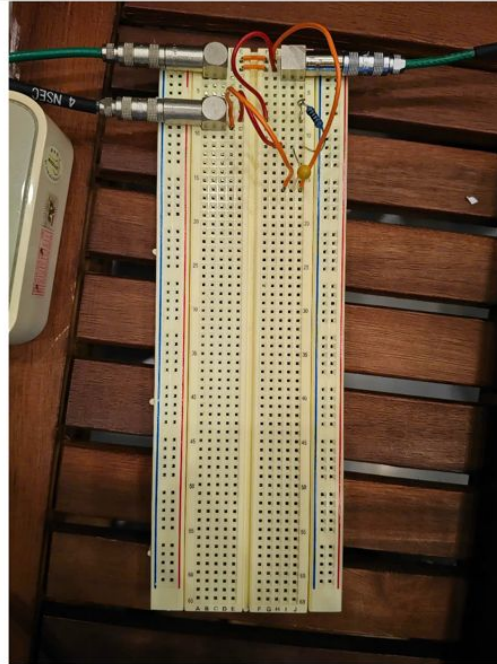
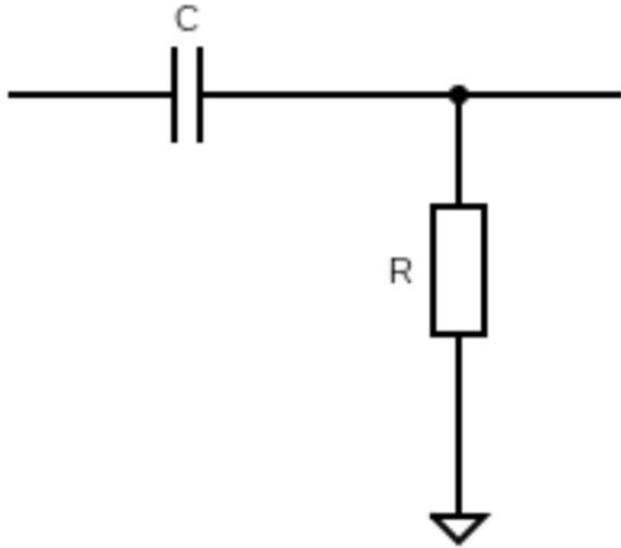
1) Theory - Low Pass Filter



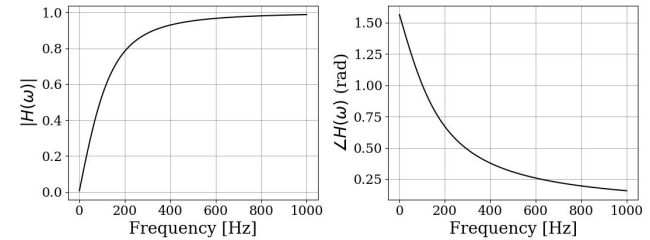
$$H(\omega) = \frac{1}{1 + i\omega RC}$$



1) Theory - High Pass Filter

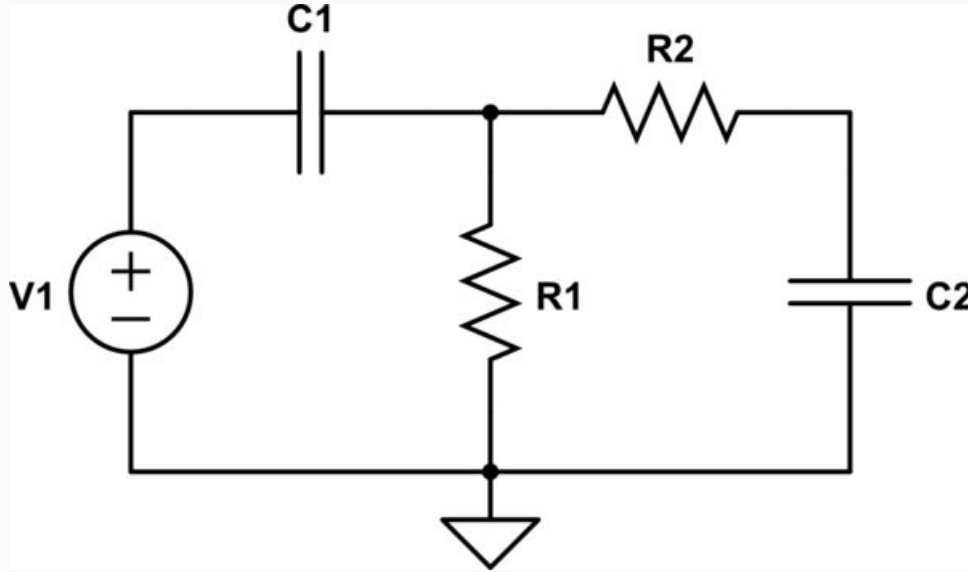


$$H(\omega) = \frac{i\omega RC}{1 + i\omega RC}$$



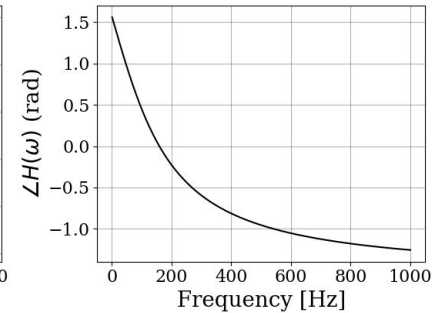
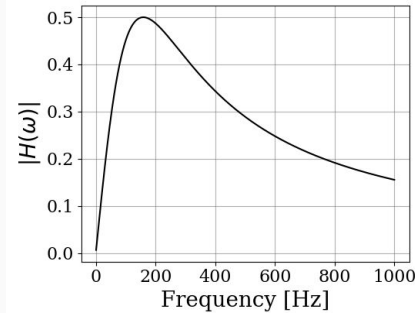
Designing a Band Pass Filter

- ❖ Combine a low pass and high pass filter:

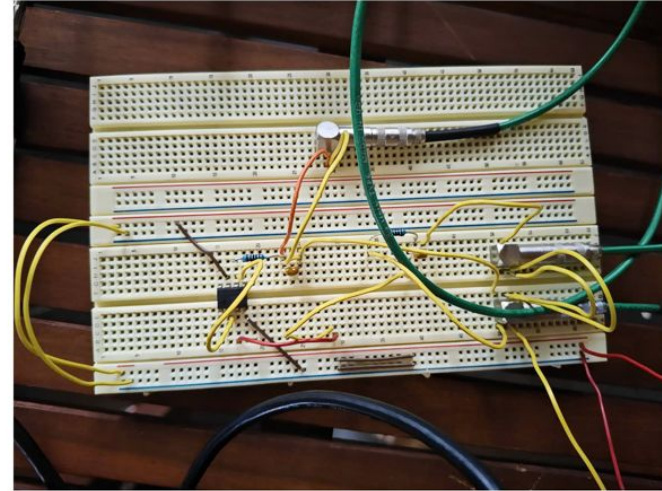
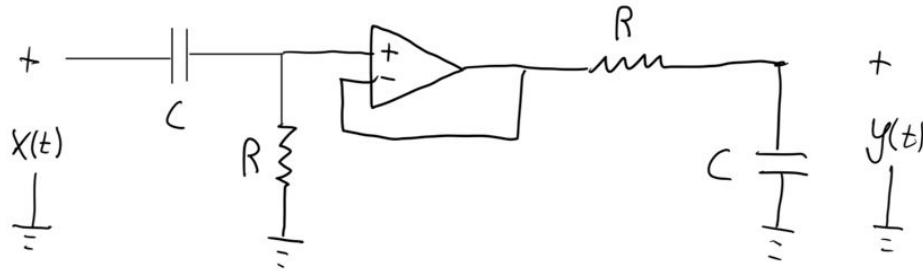


- The transfer function should be the multiplication of the low pass filter and high pass filter transfer functions:

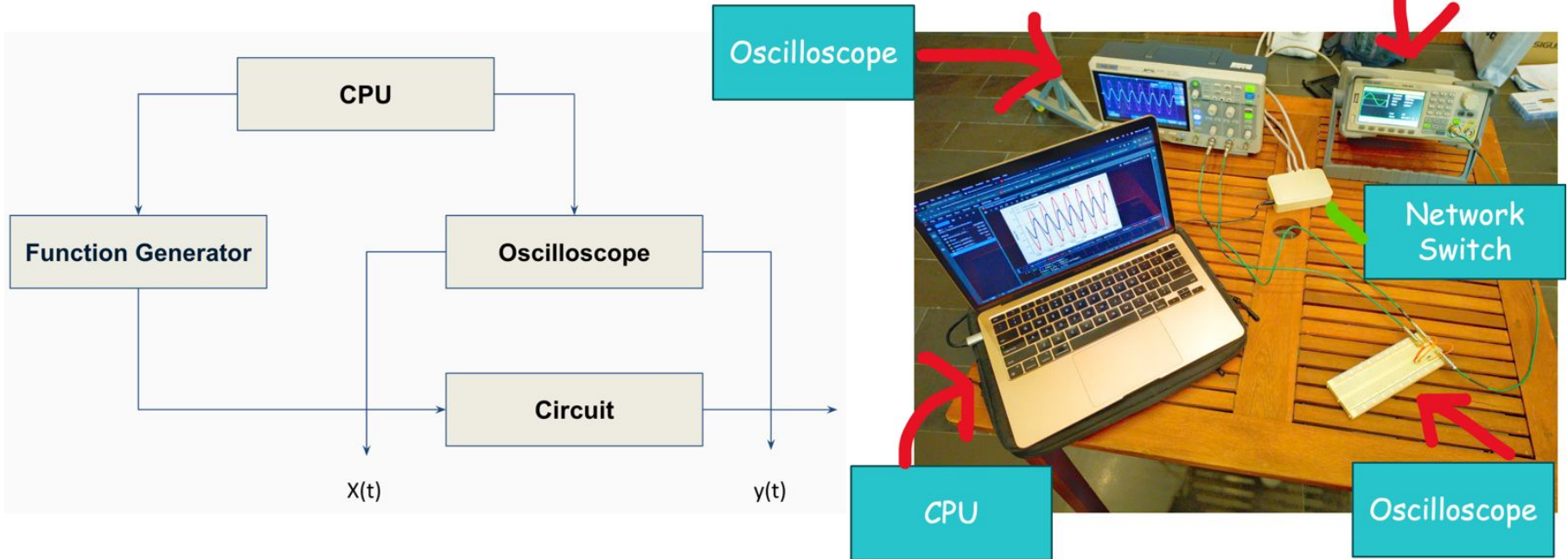
$$H(\omega)_{bp} = H(\omega)_{hp} \cdot H(\omega)_{lp} = \left(\frac{1}{1 + i\omega RC} \right) \left(\frac{i\omega RC}{1 + i\omega RC} \right)$$



1) Theory - Band Pass Filter



2) Measurement - Set up



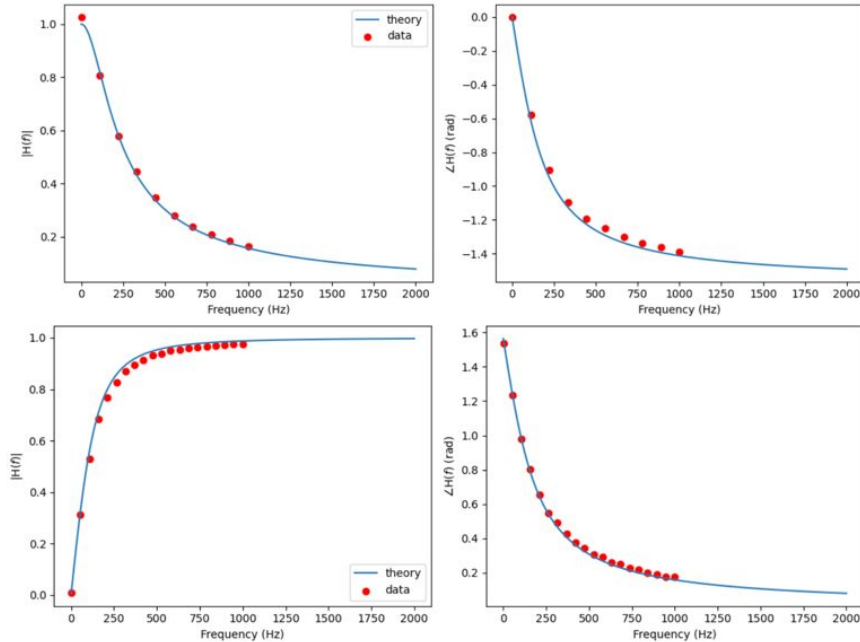
2) Measurement - Codes

```
[ ]: '''Filter testing'''
H2 = []
H3 = []
def four(time,voltage):
    T = time[1] - time[0]
    freqs, x_fft = FFT(T,voltage)
    N = freqs.shape[0]
    return freqs[0:N//2], x_fft[0:N//2]
fs = np.linspace(2, 1000, 20)
#periods = 14
for frequency in fs:
    fgen_funcs.change_frequency(fun,frequency)
    scope_funcs.set_xscale(scope,1/frequency)
    wait_time = 30*1/frequency
    print(f'Waiting for time {wait_time}')
    time.sleep(wait_time)
    T, t2, v2 = scope_funcs.capture_data(scope, 2)
    time.sleep(1)
    T, t3, v3 = scope_funcs.capture_data(scope, 3)
    f2,V2 = four(t2,v2)
    V2_mag = np.abs(V2)
    V2_phase = np.angle(V2)
    peak_index2 = np.argmax(V2_mag)
    f3,V3 = four(t3,v3)
    V3_mag = 2*np.abs(V3)
    V3_phase = np.angle(V3)
    peak_index3 = np.argmax(V3_mag)
    H2.append(np.abs(V3[peak_index3])/np.abs(V2[peak_index2]))
    H3.append(np.angle(V3[peak_index3])-np.angle(V2[peak_index2]))
```

What the codes read is:

- Step 1: Set the frequency for the function generator.
- Step 2: Capture the data from the signal.
- Step 3: Do some Fourier stuffs.
- Step 4: Take the peak amplitude value of the signal of both input and output.
- Step 5: Calculate the phase and the magnitude of the transfer function.
- Step 6: Repeat the whole things with other frequencies.

3) Results

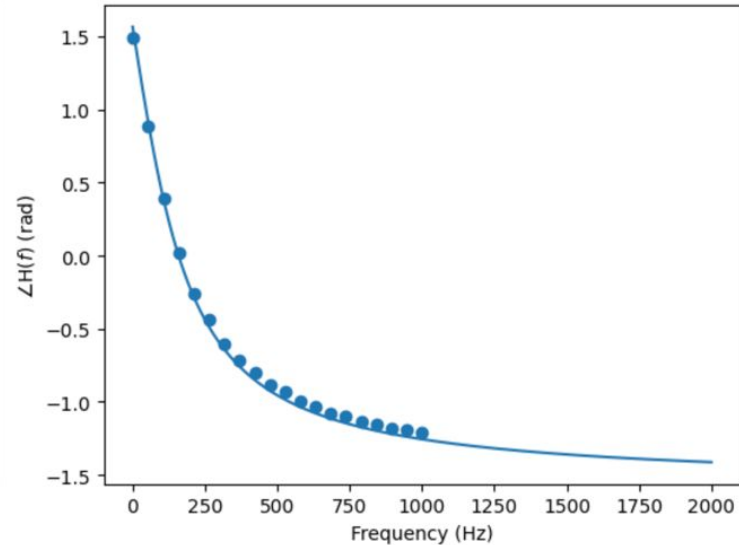
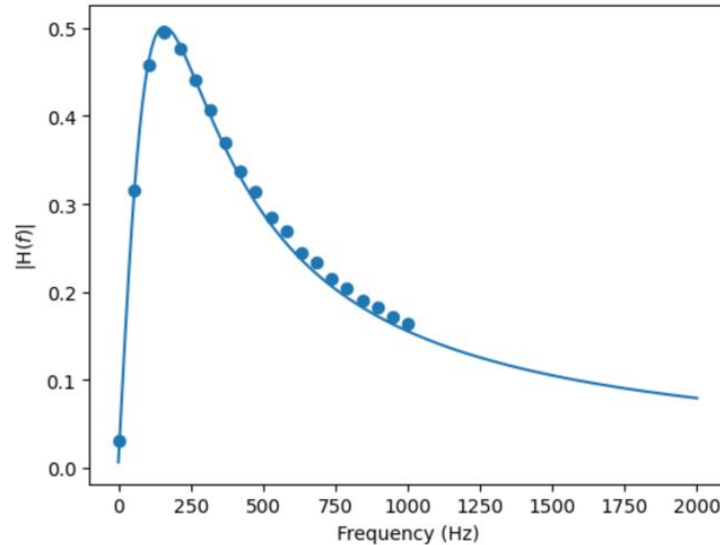


Low pass
(0-1000Hz)

High pass
(1-1000Hz)

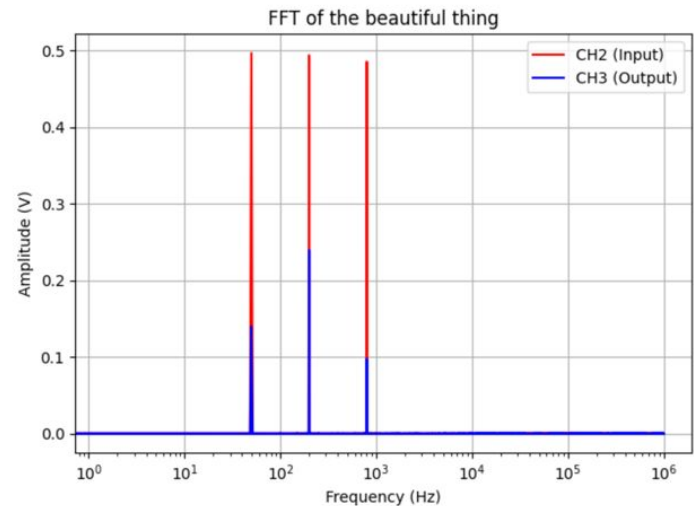
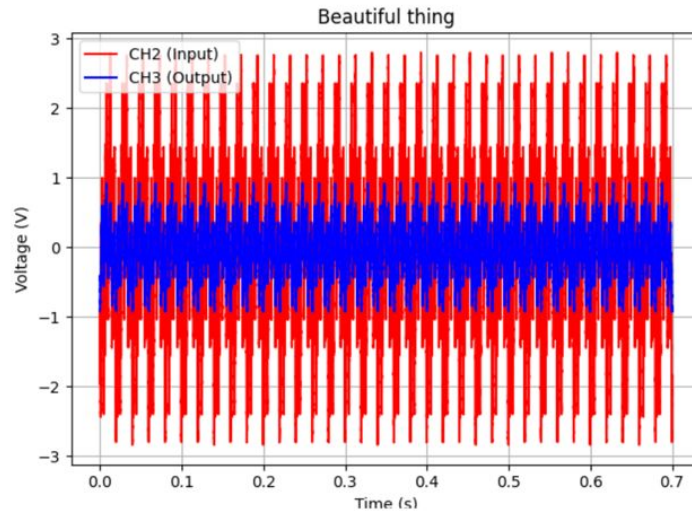
3) Results

After op-amp buffer



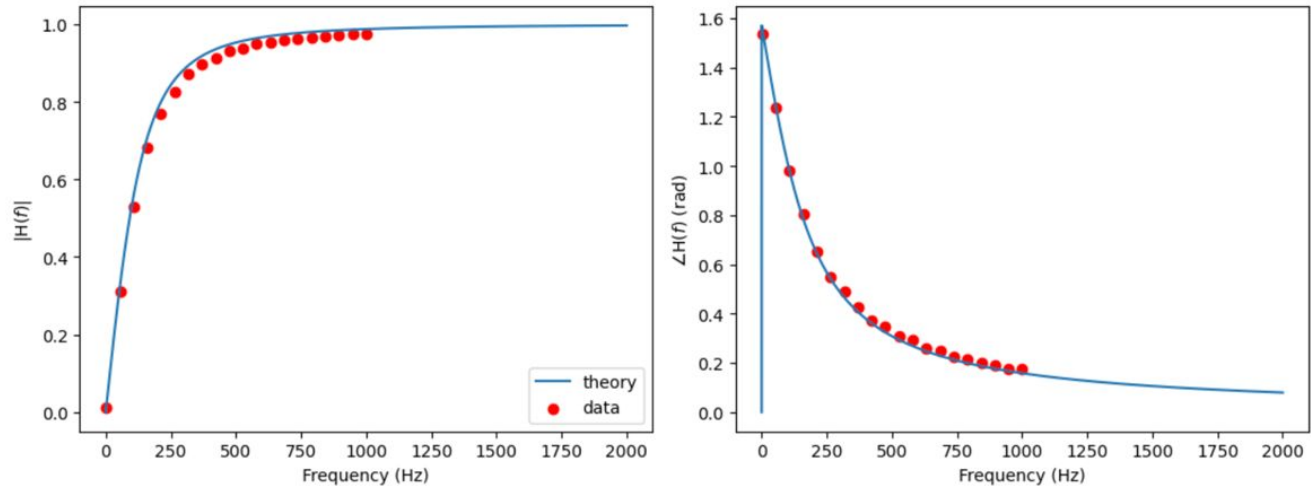
Band pass (0-1000Hz)

3) Results



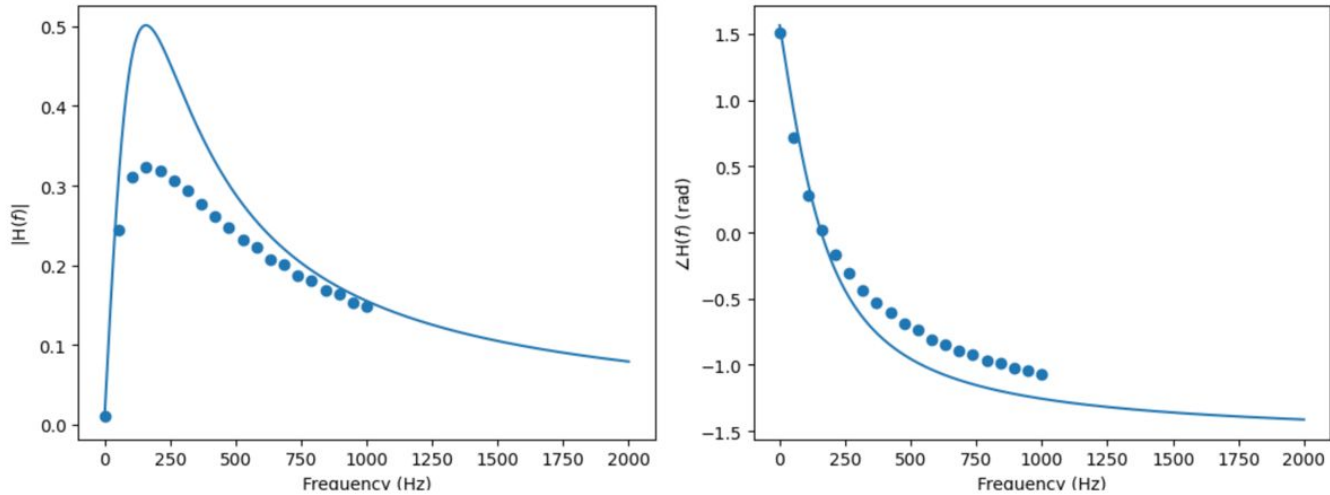
Testing the band pass filter

4) Discussion

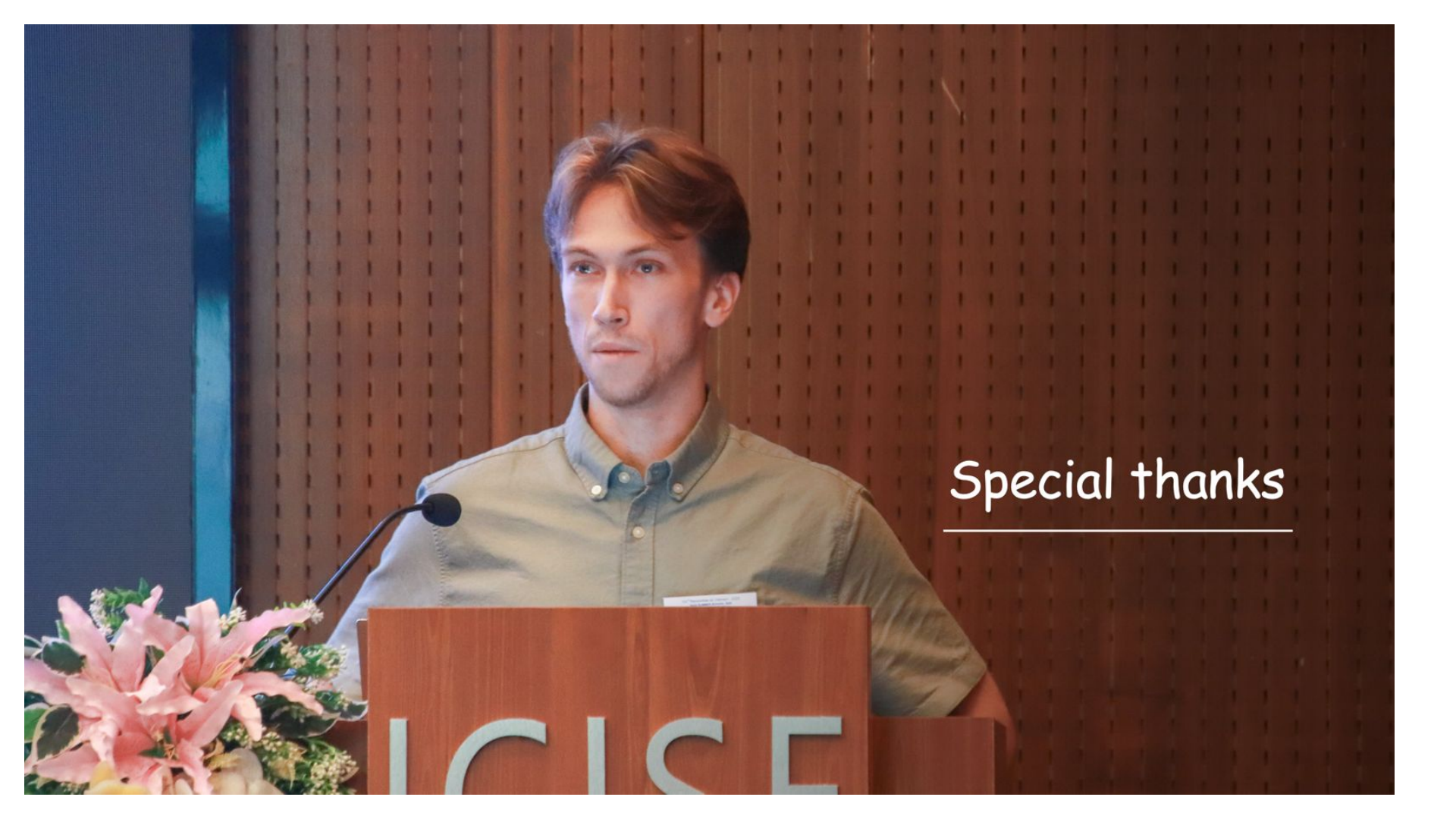


High pass (0-2000Hz)

4) Discussion



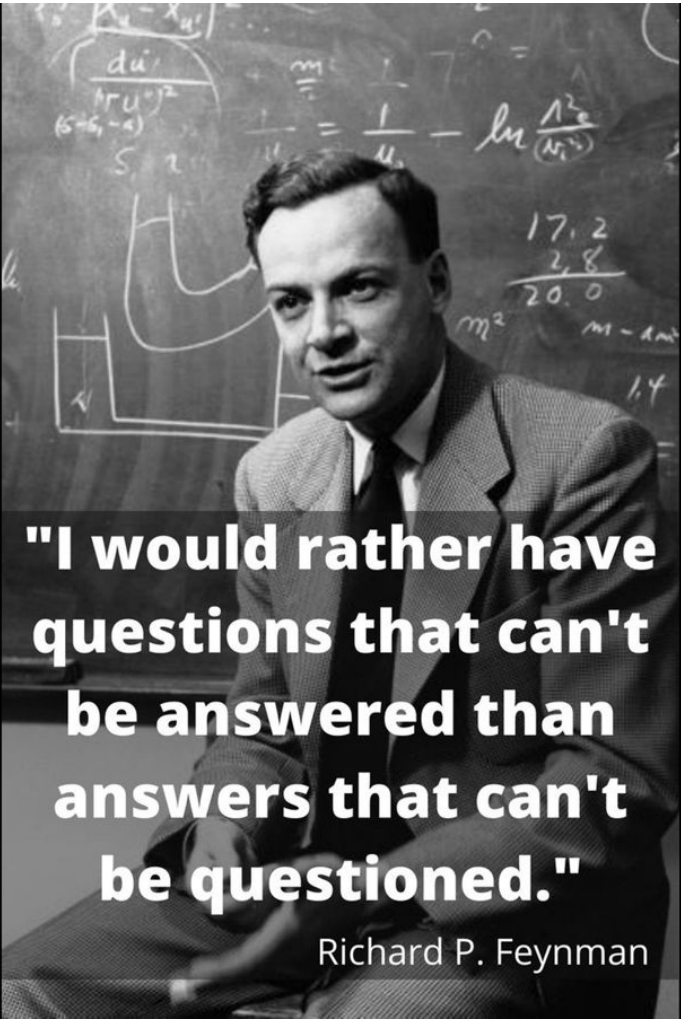
Band pass (Without buffering)



Special thanks

Thank you for looking

Do you have any
questions?

A black and white photograph of physicist Richard P. Feynman. He is seated, wearing a suit and tie, looking towards the camera with a slight smile. Behind him is a chalkboard filled with mathematical equations and diagrams. Visible equations include $\frac{du}{(5+5u-a)^2}$, $\frac{1}{u} = \frac{1}{u_0} - \ln \frac{A^2}{A_0^2}$, and a vertical calculation $\begin{array}{r} 17.2 \\ 2.8 \\ \hline 20.0 \end{array}$. There are also some handwritten notes like m^2 and $m = 1.4m$.

**"I would rather have
questions that can't
be answered than
answers that can't
be questioned."**

Richard P. Feynman